Poisson regression

Statistics III - CdL SSE

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- GLMs for **count data** are very common and have theoretical connections with binary and binomial models.
- This unit focuses on **Poisson regression models**.
- I will not cover the analysis of **contingency tables**.
- Such a topic is nonetheless discussed in the textbook but is not part of the exam.
- The most important aspects have already been covered in Unit B.

The content of this Unit is covered in **Chapter 5** of Salvan et al. (2020). Alternatively, see **Chapter 7** of Agresti (2015).

Notation and recap

- In a **Poisson regression** model, we observe Y_i independent Poisson random variables, so that

$$Y_i \overset{\text{ind}}{\sim} \text{Poisson}(\mu_i), \qquad g(\mu_i) = \eta_i = \boldsymbol{x}_i^T \beta, \qquad i = 1, \dots, n.$$

- The **canonical link** is $g(\cdot) = \log(\cdot)$, which implies a **multiplicative structure** for the mean.
- Under the canonical link, the **likelihood equations** are

$$\sum_{i=1}^{n} (y_i - \mu_i) x_{ir} = 0, \qquad r = 1, \dots, p.$$

The solution therefore has a nice interpretation as a **method of moments** estimator, in that

$$\sum_{i=1}^{n} y_i x_{ir} = \sum_{i=1}^{n} \mathbb{E}(Y_i) x_{ir}, \qquad r = 1, \dots, p.$$

Interpretation of the regression coefficients

- Under the **logarithmic link**, the mean has a multiplicative structure, namely

$$\mu_i = \exp(\boldsymbol{x}_i^T \beta) = \exp(\beta_1)^{x_{i1}} \times \cdots \times \exp(\beta_p)^{x_{ip}} = \prod_{j=1}^{p} \alpha_j^{x_{ij}}, \qquad \alpha_j = \exp(\beta_j).$$

- As a result, a **unit increase** of the jth covariate from x_{ij} to $x_{ij} + 1$, keeping the other covariates fixed, has the following impact on the new mean, say μ_{new}

$$\mu_{\text{new}} = \alpha_1^{x_{i1}} \times \cdots \times \alpha_j^{x_{ij}+1} \times \cdots \times \alpha_p^{x_{ip}} = \alpha_j \left(\alpha_1^{x_{i1}} \times \cdots \times \alpha_p^{x_{ip}}\right) = \alpha_j \mu_i.$$

  In other words, the regression parameters, once exponentiated, can be interpreted as **relative changes** of the mean, namely

$$\alpha_j - 1 = \exp(\beta_j) - 1 = \frac{\mu_{\text{new}} - \mu_i}{\mu_i}.$$

> The interpretation in terms of relative changes is a consequence of the **logarithmic link** function. Therefore, the same interpretation applies whenever this link is used.

Exposure rate

- Often the expected value of a response count Y_i is proportional to an index t_i, the **exposure**. For instance, t_i might be an amount of time and/or a population size, such as in modeling crime counts for various cities. Or, it might be a spatial area, such as in modeling counts of plant species.
- In these cases, the **sample rate** is Y_i/t_i, with expected value μ_i/t_i. With explanatory variables, a model for the expected rate under a **logarithmic link** has the form

$$\log\left(\frac{\mu_i}{t_i}\right) = \boldsymbol{x}_i^T\beta \qquad \Longrightarrow \qquad \log\mu_i = \boldsymbol{x}_i^T\beta + \log t_i \qquad \Longrightarrow \qquad \mu_i = t_i\exp(\boldsymbol{x}_i^T\beta).$$

- The adjustment term $\log t_i$ is called an **offset**. The fit corresponds to using $\log t_i$ as an **explanatory variable** and **forcing its coefficient** to equal 1. In **R**, it is specified as `offset(log(t))` in the model formula.
- The mean is proportional to t_i, with a proportionality constant $\exp(\boldsymbol{x}_i^T\beta)$ that depends on the covariates. The quantities $\exp(\beta_j)$ are interpreted as **relative changes** of the **rate**.
- Whether the coefficient of $\log t_i$ is indeed equal to 1 can be checked by estimating it as a free parameter, as in the `ships` case study below.

Overdispersion

- In Poisson regression the main **assumption** is that $Y_i \sim \text{Poisson}(\mu_i)$, implying that

$$\text{var}(Y_i) = \mu_i,$$

  where implicitly we have set $\phi = 1$.

- However, from the analysis of the residuals or by comparing the Pearson X^2 statistic with $n - p$, we may realize that the data present **overdispersion**, namely that $\text{var}(Y_i) > \mu_i$. A **common specification** is

$$\text{var}(Y_i) = \phi\mu_i,$$

  with $\phi > 1$. In either case, the Poisson regression model is **misspecified**.

- The two most common solutions to overdispersion are the following:

  i. the use of **quasi-likelihoods**, discussed in Unit E;

  ii. the use of another parametric distribution; a typical choice is the **negative binomial**.

Zero-inflation

- The frequency of **zero outcomes** is often **larger than expected** under a Poisson model. Since the Poisson **mode** is $\lfloor \mu \rfloor$ (for non-integer μ), a Poisson GLM is inadequate when the mean is large but the modal response is 0.
- Such data are called **zero-inflated**: some subjects have a **structural zero**, while the others have positive counts or **random zeros**.
- Example: weekly gym visits. Some people never exercise (structural zero), others exercise occasionally but not in the past week (random zero).
- The two most common solutions to zero-inflation are the following:
  i. the **zero-inflated Poisson** (ZIP) model, a mixture of a point mass at zero, with weight π_i, and a $\text{Poisson}(\mu_i)$, so that $\mathbb{P}(Y_i = 0) = \pi_i + (1 - \pi_i)e^{-\mu_i}$;
  ii. **hurdle models**, which model zero vs. nonzero outcomes and then the positive counts through a zero-truncated distribution.

Case study: the `ships` data

The `ships` dataset

- The `ships` dataset of the `MASS` library, analysed in McCullagh and Nelder (1989) (Section 6.3.2), concerns a type of **damage** caused by waves to the forward section of cargo ships.
- The data are grouped into 40 cells. The variables are:
  - `type`: ship type, A to E;
  - `year`: year of construction, 1960–64, 1965–69, 1970–74, 1975–79 (coded as 60, 65, 70, 75);
  - `period`: period of operation, 1960–74 or 1975–79 (coded as 60, 75);
  - `service`: aggregate **months of service**;
  - `incidents`: **number of damage incidents**.

	type	year	period	service	incidents
1	A	60	60	127	0
2	A	60	75	63	0
3	A	65	60	1095	3
4	A	65	75	1095	4
5	A	70	60	1512	6
6	A	70	75	3353	18

> This case study is based on the exam of 29 April 2015.

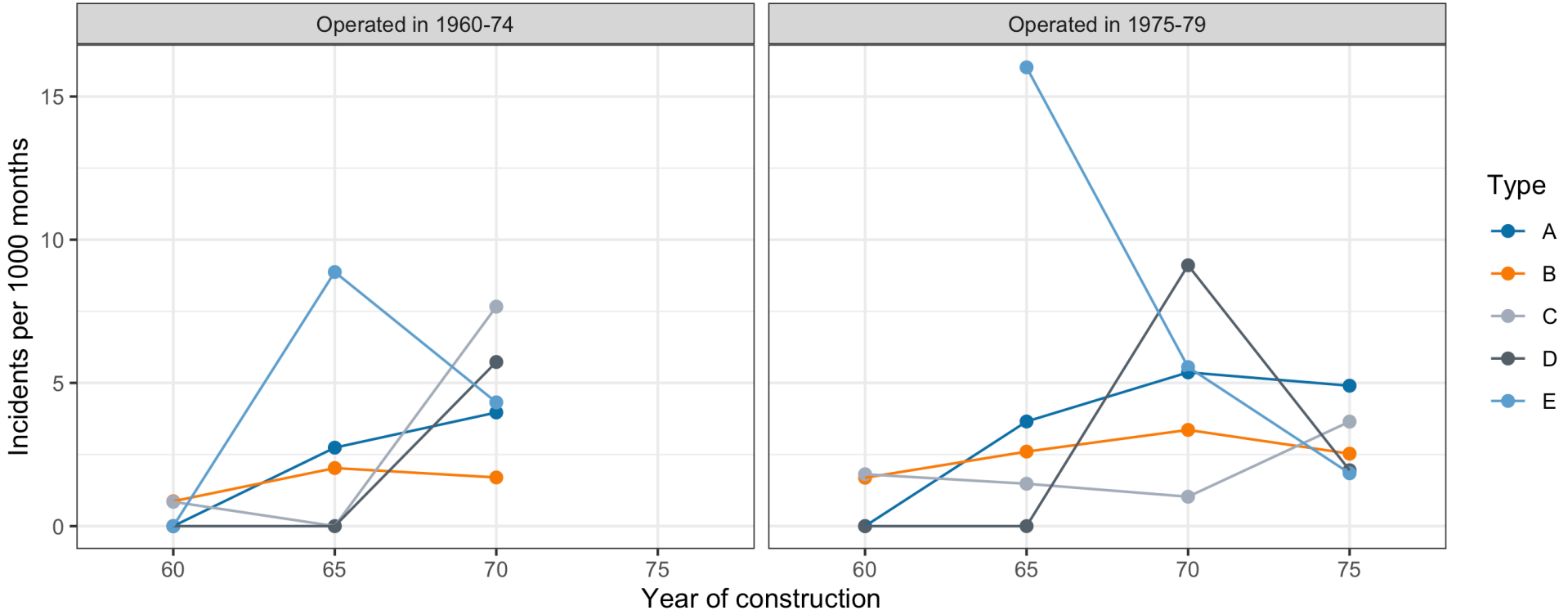
Ships that never sailed

- There are 6 cells with `service` $= 0$ and therefore, necessarily, `incidents` $= 0$:

	type	year	period	service	incidents
7	A	75	60	0	0
15	B	75	60	0	0
23	C	75	60	0	0
31	D	75	60	0	0
34	E	60	75	0	0
39	E	75	60	0	0

- Five of them are ships built in 1975–79 that would have operated in 1960–74, which is **impossible**.
- These cells carry **no information**: if $t_i = 0$, then $\mu_i = t_i \exp(\boldsymbol{x}_i^T \beta) = 0$ whatever the value of β. We **remove** them, leaving $n = 34$ cells.

Exploratory analysis



- The `service` ranges from 45 to 44,882 months: the counts are not comparable, and the **rates** must be considered instead.
- Rates tend to be higher for ships **built after 1964** and in the **period 1975–79**, and they vary across types. Cells with little service, mostly of types D and E, give very **noisy** rates, often equal to zero.

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Modelling the `ships` data

- Let Y_i be the number of incidents and t_i the months of service of the ith cell. A linear model would be inappropriate: the Y_i are **counts** whose variability grows with the mean, and they depend on the **exposure** t_i.
- We consider the Poisson **rate model**

$$Y_i \overset{\text{ind}}{\sim} \text{Poisson}(\mu_i), \qquad \log \mu_i = \log t_i + \boldsymbol{x}_i^T \beta, \qquad i = 1, \ldots, 34,$$

  where $\boldsymbol{x}_i$ contains the intercept and the dummy variables of `type`, `year` and `period`, so that $p = 9$.
- The **baseline** is a ship of type A, built in 1960–64 and operated in 1960–74.
- Hence, $\exp(\boldsymbol{x}_i^T \beta) = \mu_i / t_i$ is the expected number of incidents **per month of service**.

The rate model

```
Call:
glm(formula = incidents ~ type + year + period + offset(log(service)),
    family = poisson, data = ships)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -6.40590    0.21744 -29.460  < 2e-16 ***
typeB       -0.54334    0.17759  -3.060  0.00222 **
typeC       -0.68740    0.32904  -2.089  0.03670 *
typeD       -0.07596    0.29058  -0.261  0.79377
typeE        0.32558    0.23588   1.380  0.16750
year65       0.69714    0.14964   4.659 3.18e-06 ***
year70       0.81843    0.16977   4.821 1.43e-06 ***
year75       0.45343    0.23317   1.945  0.05182 .
period75     0.38447    0.11827   3.251  0.00115 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- The **likelihood ratio test** against the null model gives 107.63 on 8 degrees of freedom: the covariates are **jointly significant**.

Is the offset appropriate?

- We estimate the coefficient of $\log t_i$ as a **free parameter**, instead of fixing it to 1:

```
  Estimate Std. Error
     0.903      0.102
```

```
Analysis of Deviance Table

Model 1: incidents ~ type + year + period + offset(log(service))
Model 2: incidents ~ type + year + period + log(service)
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        25     38.695
2        24     37.804  1  0.89079   0.3453
```

- The estimate is close to 1 and the likelihood ratio test does not reject H_0: the coefficient equals 1. The **rate model** is supported by the data.

Interpretation of the coefficients

	$\exp(\hat{\beta}_j)$	Lower 95%	Upper 95%
typeB	0.581	0.410	0.823
typeC	0.503	0.264	0.958
typeD	0.927	0.524	1.638
typeE	1.385	0.872	2.199
year65	2.008	1.498	2.692
year70	2.267	1.625	3.162
year75	1.574	0.996	2.485
period75	1.469	1.165	1.852

- The estimated rate for the baseline is $1000 \exp(\hat{\beta}_1) = 1.65$ incidents per 1000 months.
- For a ship of **type B**, the rate is multiplied by 0.581, i.e. it is 41.9% **lower** than for type A, the other covariates being equal.
- In the **period 1975–79**, the rate is 46.9% **higher** than in 1960–74. Ships **built in 1965–74** have about **twice** the rate of those built in 1960–64.

Goodness of fit and overdispersion

- The deviance is 38.7 on 25 degrees of freedom (p-value 0.04) and $X^2 = 42.28$, so that $\hat{\phi} = 1.69$: there is a **moderate overdispersion**.
- Using a **quasi-Poisson** model (Unit E), the estimates are unchanged, while the standard errors are multiplied by $\sqrt{\hat{\phi}} = 1.3$:

```
Single term deletions

Model:
incidents ~ type + year + period + offset(log(service))
       Df Deviance F value   Pr(>F)
<none>      38.695
type    4   62.365  3.8232 0.014728 *
year    3   70.103  6.7640 0.001706 **
period  1   49.355  6.8873 0.014590 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- `type`, `year` and `period` remain **significant** at the 5% level.

Conclusions

- The number of incidents must be modeled relative to the **months of service**: the offset $\log t_i$ is supported by the data.
- The incident rate depends on the **type** of ship, its **year of construction** and the **period of operation**.
- There is a **moderate overdispersion**. Accounting for it widens the confidence intervals, but the conclusions do not change.

References

Agresti, A. (2015), *Foundations of Linear and Generalized Linear Models*, Wiley.

McCullagh, P., and Nelder, J. A. (1989), *Generalized linear models*, Chapman & Hall/CRC.

Salvan, A., Sartori, N., and Pace, L. (2020), *Modelli lineari generalizzati*, Springer.